ModelIt!

Student Activity Pack

Why Did Dinosaurs Rule and Then Vanish?

Modeling Extinction Dynamics and the Fragility of Dominant Species

Grade 10L2 | NGSS HS-LS4-1, HS-LS4-4 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think caused the extinction of the dinosaurs — and why didn't ALL life go extinct?

2 Why might being the biggest and strongest animal actually be a disadvantage during a catastrophe?

3 Draw a diagram showing what you think happens to an ecosystem when its dominant species suddenly disappears.

4 What traits do you think helped some species survive the asteroid impact that killed the dinosaurs?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Species Diversity | Environmental Change Rate | Adaptation Speed | Extinction Pressure | Niche Competition | Survival Advantage | Population Recovery

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Gradual Change

Slow environmental shift over millions of years

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Asteroid Impact

Sudden catastrophic environmental change

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Post-Extinction

Recovery after mass extinction

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Food Web Complexity — The number and strength of feeding relationships in an ecosystem, which determines how disruption of one species cascades through the entire community

Genetic Diversity — The range of genetic variation within a species' population, which determines the raw material available for natural selection and the potential for rapid adaptation

Geographic Range — The total area occupied by a species, which affects vulnerability — widespread species are less likely to be completely eliminated by a localized catastrophe

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Dinosaurs dominated Earth for 165 million years — then vanished almost overnight. What determines whether a species survives or goes extinct when the environment changes catastrophically? Use evidence from your simulation to support your answers.

1. **How does your model explain why 165 million years of evolutionary success provided no protection against the asteroid impact?**

2. **What does your model reveal about the relationship between Adaptation Speed and Environmental Change Rate — why is there a critical threshold?**

3. **Why did small body size become a Survival Advantage during the mass extinction when large body size was advantageous before it?**

4. **How does your model's Population Recovery phase relate to the concept of adaptive radiation?**

5. **What parallels does your model suggest between the Cretaceous mass extinction and the current biodiversity crisis driven by human activity?**

8 STEM Challenge: Design a Biodiversity Resilience Index

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YOUR ENGINEERING MISSION

Design a scoring system that evaluates how vulnerable or resilient different species are to rapid environmental change, based on their biological traits and ecological relationships.

THE SCENARIO

Conservation biologists need a tool that predicts which modern species are most vulnerable to rapid environmental change (climate change, habitat destruction, pollution). Your team has been hired to design a Biodiversity Resilience Index that scores species based on the same survival factors your model identified.

Engineering Design Thinking

1. WHAT DATA WOULD YOU NEED TO CALCULATE A RESILIENCE SCORE FOR ANY GIVEN SPECIES?
2. HOW WOULD YOU ACCOUNT FOR SPECIES THAT ARE SPECIALISTS VERSUS GENERALISTS?
3. WHAT THRESHOLD SCORE WOULD INDICATE A SPECIES IS AT HIGH EXTINCTION RISK?
4. HOW WOULD YOUR INDEX HANDLE INTERCONNECTED SPECIES — IF A KEYSTONE SPECIES SCORES LOW, HOW DOES THAT AFFECT DEPENDENT SPECIES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Paleobiologists study ancient life through fossils to understand evolution, extinction, and the history of biodiversity. They work at museums, universities, and research institutions, earning $55,000–$120,000/year. Conservation Biologists who apply extinction science to modern species protection earn $50,000–$110,000/year.